YAKEEN

NEET 2.0 2027

Physics | Lecture - 10

Basic Maths and Calculus

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Partial Differentiation

Partial differentiation

$V(x, y, z)$ = 'V' is a function of x, y & z

08:50

$$\left(\frac{dV}{dx}\right)_{y \& z = \text{const}}$$

$$\left(\frac{dV}{dy}\right)_{x, z = \text{const.}}$$

$$\left(\frac{dV}{dz}\right)_{y \& x = \text{const}}$$

08:52

(8) $V = x^2 y^3 z^4$ find $\left(\frac{\partial v}{\partial x}\right) = \frac{\partial (x^2 y^3 z^4)}{\partial x} = y^3 z^4 \frac{\partial x^2}{\partial x}$

Partial diff^n

$= 2x y^3 z^4$

$\left(\frac{\partial v}{\partial y}\right) = \frac{\partial (x^2 y^3 z^4)}{\partial y} = x^2 z^4 (3y^2)$

$\frac{\delta v}{\partial z} = x^2 y^3 \frac{dz^4}{dz} = 4z^3 x^2 y^3$

(0) gf electric Potential $V = x^2y + y^2z + z^2x$ find

Electric field $\vec{E} = -\left[\frac{dV}{dx}\hat{i} + \frac{dV}{dy}\hat{j} + \frac{dV}{dz}\hat{k}\right]$.

09:01

NEET/JEE may time ✳

Soln

$$\# \left(\frac{dV}{dx}\right)_{y\,\&\,z\ \text{Cnst.}} = \frac{d\left(x^2y + (y^2z) + z^2x\right)}{dx} = y\frac{dx^2}{dx} + \frac{d(y^2z)}{dx} + z^2\frac{dx}{dx}$$

$$= y\,2x + 0 + z^2 \times 1$$

$$\# \left(\frac{dV}{dy}\right)_{x\,\&\,z=\text{Cnst}} = x^2 \times 1 + z(2y) + 0 \qquad = (2xy + z^2)$$

$$= 2yz + (x^2)$$

$$\left(\frac{dV}{dz}\right)_{xy\ \text{Cnst.}} = 0 + y^2\frac{dz}{dz} + x\frac{dz^2}{dz} = (y^2 + 2xz)$$

If $V = x^3z + y^3x + xyz$

find $\frac{dv}{dx}\hat{i} + \frac{dv}{dy}\hat{j} + \frac{dv}{dz}\hat{k}$

$\frac{dv}{dx} =$

$\frac{dv}{dy} =$

$\frac{dv}{dz} =$

$$y = e^{(-x)}$$

$$\frac{y}{x} = \frac{d(e^{-x})}{dx} = e^{-x} \times \frac{d(-x)}{dx}$$

$$= -e^{-x} \frac{dx}{dx}$$

$$= -e^{-x}$$

$$y = e^{-x^2}$$

$$\frac{dy}{dx} = e^{-x^2} \times \frac{d(-x^2)}{dx}$$

$$= e^{-x^2}[-2x]$$

$$= -2x\, e^{-x^2}$$

$y = A\cos(Kx)$

$\frac{dy}{dx} = -A\sin(Kx)\frac{d(Kx)}{dx}$

$= -AK\sin(Kx)$

$y = 5e^{3x}$

$\frac{dy}{dx} = 5e^{3x} \times 3$

$= 15e^{3x}$

$y = e^{-4x}$

$\frac{dy}{dx} = e^{-4x}(-4)$

$= -4e^{-4x}$

$y = 3\sin(x^2+2)$

$\frac{dy}{dx} = 3\cos(x^2+2) \times [2x+$

$= 6x\cos(x^2+2)$

09:09

(b) $q = q_0 e^{-\alpha t^2}$

find $\frac{dq}{dt}$ where q_0 & α are cosⁿ.

$$\frac{dq}{dt} = q_0 \frac{d e^{-\alpha t^2}}{dt}$$

$$= q_0 e^{-\alpha t^2} \times (-\alpha 2t)$$

$$= -q_0 \alpha 2t \left[e^{-\alpha t^2}\right]$$

(a) $q = q_0\left(1 - e^{\frac{\alpha}{\beta} t}\right)$ find $\frac{dq}{dt}$

$$\frac{dq}{dt} = q_0 \left[\frac{d(1)}{dt} - \frac{d\left(e^{+\alpha/\beta t}\right)}{dt}\right]$$

$$= q_0 \left[0 - e^{\alpha/\beta t} \times \frac{\alpha}{\beta}(1)\right]$$

$$= -\frac{\alpha q_0}{\beta} e^{\left(\frac{\alpha}{\beta}\right)t}$$

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(Q) If $y = 2t^3 + 1$ and $x = t^2$ then find $\left(\frac{dy}{dx}\right) = ??$

Soln

$$\frac{dy}{dt} = 2(3t^2) + 0$$

$$= 6t^2 \quad -(I)$$

$$\frac{dx}{dt} = 2t \quad -(II)$$

$(I)/(II)$

$$\frac{dy/dt}{dx/dt} = \frac{6t^2}{2t}$$

$$\left[\frac{dy}{dx}\right] = 3t$$

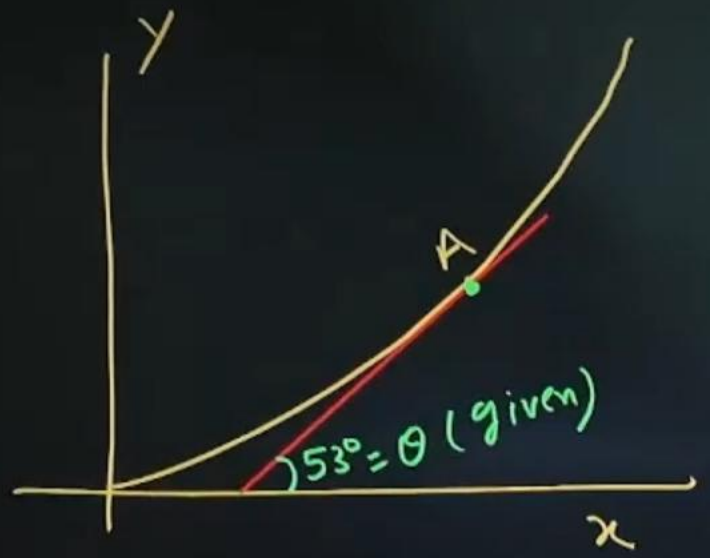
(a) If $y = \sin\theta$ and $x = \cos\theta$ find $\frac{dy}{dx}$

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$$\frac{dy}{d\theta} = \cos\theta$$

$$\frac{dx}{d\theta} = -\sin\theta$$

$$\frac{dy}{dx} = -\cot\theta$$



find slope at 'A'

slope = $\tan 53°$

$m. = \frac{4}{3}$

(θ) find slope of a curve $y = 2x^2$ at $x = 1$.

$$\text{slope} = \frac{dy}{dx}$$

$$= \frac{d(2x^2)}{dx} = 2(2x)$$

$$= 4(x)$$

$$= 4(1)$$

$$\text{Slope} = \frac{dy}{dx} = 4$$

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(a) find slope of equation $y = 3x^2 + 5x - 4$ at $x = 1$

09:25

$$\frac{dy}{dx} = 3(2x) + 5(1) - 0$$

$$\left|\frac{dy}{dx}\right|_{x=1} = 6x + 5$$

$$= 6 \times 1 + 5$$

$$= 11$$

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If $y = \frac{4}{(x^2-5)}$ find $\frac{dy}{dx} = ??$

outside In

$y = 4(x^2-5)^{-1}$

$\frac{dy}{dx} = -1 \times 4[x^2-5]^{-2} \times [2x]$

$$\boxed{\frac{dy}{dx} = \frac{-8x}{(x^2-5)^{+2}}}$$

division Rule

$$\frac{dy}{dx} = \frac{0[x^2-5] - 4[2x-0]}{(x^2-5)^2}$$

$$\boxed{\frac{dy}{dx} = \frac{-8x}{(x^2-5)^2}}$$

$$y = \frac{3}{\sqrt{x^2-4x}} = 3(x^2-4x)^{-\frac{1}{2}}$$

Soln $\frac{dy}{dx} = 3\left(-\frac{1}{2}(x^2-4x)^{-\frac{1}{2}-1}\right] \times$

$= -\frac{3}{2}(x^2-4x)^{-3/2}[2x-4]$

Position

$\frac{dx}{dt}$ = The rate of change in 'x' w.r.t time = velocity

$\frac{d^2y}{dx^2}$ = Rate of change in $\left(\frac{dy}{dx}\right)$ w.r.t x

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Area

$\frac{dA}{dt}$ = Rate of change in Area w.r.t time

momentum

$\frac{dP}{dt}$ = The rate of change in momentum w.r.t time = force

velocity

$\# \frac{d\vec{v}}{dt}$ = Rate of change in velocity w.r.t time = accele

speed

$\# \frac{d|\vec{v}|}{dt}$ = Rate of change in magnitude of velocity w.r.t time

= diff^n of speed w.r.t time

= Tangential acc^n.

$\left|\frac{dy}{dx}\right|$ = Magnitude of Rate of change in y w.r.t x

$\frac{d(y^2)}{dx}$ = Rate of change in $[y^2]$ w.r.t x

...e of change in magnitu... of "y" w.r.t x

$\frac{dP}{dt}$

$\left|\frac{dP}{dt}\right|$

$\frac{d|P|}{dt}$

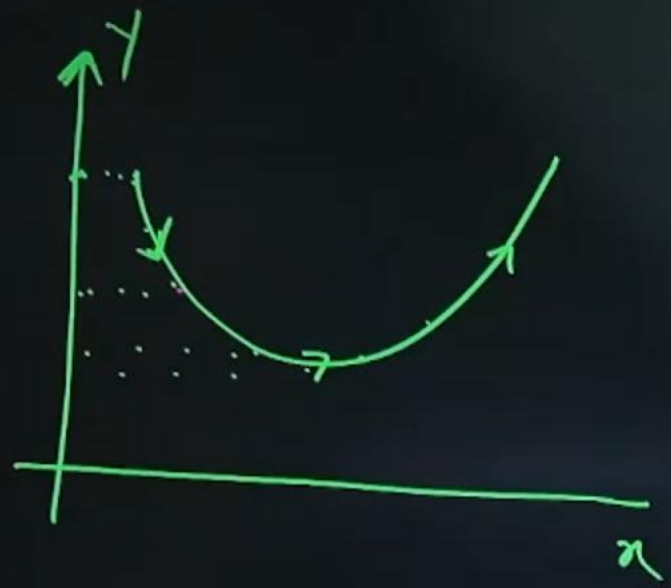
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$\# \quad \frac{d^2y}{dx^2}$ = The rate of change in $\left(\frac{dy}{dx}\right)$ w.r.t x

= The rate of change in (slope) w.r.t x

$\Rightarrow \quad \frac{dy}{dx}$ = The rate of change in y w.r.t. x

= slope



⇒ 1st y is dec ↓ then y is ↑ (Inc)

⇒ Slope $\left[\frac{dy}{dx}\right]$ → continously Increasing

⇒ $\frac{d^2y}{dx^2}$ = Rate of change in slope $\left(\frac{dy}{dx}\right)$

Thank You